Date. No. 5-dars

L1 va L2 Regularization (balanslashtirish)

- L1 Regularization (Lasso Regression)
  - Absolute value as a penalty (for features)
  - Muhim featurelar muhimligi
  - Bazi featurelarni tashlab yuboradi
- Analogy
  - Sumkamizda joy yoq shunda muhim bolmagani qoladi

(Ridge Regression)

- L2 Regularization (Ridge Regression) (overfittingda yordam beradi)
  - Squared value as a penalty
  - Bazi featurelarni tasirini kamaytiradi lekin tashlab yubormaydi
- Analogy
  - Sumkamizda joy yoq lekin barcha narsalar muhim (Mr Bean)

shunda compressor orqali ularni siqamiz va barchasini olib ketamiz

$$L_1 = \lambda * \sum |\beta_i|$$

λ → Parametr

$$L_2 = \lambda * \sum \beta_i^2$$

Lasso Regression

- Linear Regression + L_1 Regularization
- Uy prediction:
  - 100 ta feature bor
  - Faqat 10 tasi muhim

Analiy Lasso Regression (L_1)

```
from sklearn.linear_model import Lasso
    lasso = Lasso(alpha = 10) => (alpha = 0.1) => (alpha = 100)
x = df.drop('num', axis = 1)
    y = df['num']
from sklearn.model_selection import train_test_split
  lasso
  lasso.fit(x_train, y_train)
  y_pred = lasso.predict(x_test)
    from sklearn.metrics import r2_score, mean_absolute_error
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
print(r2)                print(mae)
```

Analiy Ridge Regression (L_2)

```
from sklearn.linear_model import Ridge
    ridge = Ridge(alpha = 10) => (alpha = 0.1) => (alpha = 100)
x = df.drop('num', axis = 1)
    y = df['num']
from sklearn.model_selection import train_test_split
  ridge
  ridge.fit(x_train, y_train)
  y_pred = ridge.predict(x_test)
    from sklearn.metrics import r2_score, mean_absolute_error
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
print(r2)                print(mae)
```

Date. No. Nimalar organdik:

ML

- Model Programming (basics)
- Algorithms:
  - Losso
  - Ridge
- Kutubxonalar
  - Matplotlib
  - Seaborn
  - Plotly express
  - Klib
  - Autoclean
  - OS
- Project 1

Python

- Built-in function